Practice Quiz: Planning / Complete Agent (Implementation)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. The CompleteActiveInferenceAgent integrates: A) Only perception B) Perception, action, learning, and habit formation in one unified class C) Only reinforcement learning D) External API calls

2. Habit formation in the implementation is achieved by: A) Hard-coding preferred actions B) Strengthening the policy prior for successful policies via `update_habits` C) Removing all but one policy D) Increasing gamma

3. The equation `log_pi = -γG + log(policy_prior)` shows that action selection depends on: A) Only habits B) Both EFE evaluation (G) and habitual tendencies (policy_prior) C) Only EFE D) Random noise

4. Over many time steps, habit entropy: A) Increases B) Decreases as certain policies become more habitual C) Stays constant D) Oscillates

5. The `step` method performs these operations in order: A) Act, Perceive, Learn, Habituate B) Perceive → Decide → Learn → Habituate C) Learn, Act, Perceive D) Habituate, then everything else

6. The `summary` method reports: A) Only action counts B) Action distribution, final beliefs, learning rate, strongest habit, and habit entropy C) The agent's name D) Memory usage

7. Comparing early vs. late behavior reveals: A) No difference B) Early behavior is goal-directed (EFE-dominated), late behavior is habitual (prior-dominated) C) Late behavior is always random D) Early behavior is always habitual

Part B: Short Answer

1. Trace the `step` method for one time step. List each operation, what it updates, and why.

2. How would you implement "dehabituation" — reducing habit strength when the environment changes? Describe the code changes needed and when dehabituation should trigger.

3. Reflecting on the entire 101 course: you've studied Active Inference through Cognitive Science, Computational Neuroscience, Mathematical Frameworks, and now Implementation. Identify one concept that became clearer through implementation and explain why.